Samarjeet Mohite

Software Engineer – Backend & AI Systems
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Summary

Backend-leaning full-stack software engineer with **2+ years** of experience, building **agentic AI systems** and the **cloud infrastructure** behind them for enterprise clients. Also worked as a **forward deployed engineer**, working directly with clients. I like owning things end to end and enjoy learning and solving the problems that come while **scaling systems**.

Education

Indian Institute of Information Technology and Management Gwalior — **Jul 2020 – May 2025**
Integrated B.Tech and M.Tech in Information Technology — *CGPA: 7.9*

Experience

Meraki Labs – Venture Studio — **Mar 2025 – Present**
Software Engineer – shipping AI-native products across ventures — *Bangalore, India*

***Nurix AI** – Voice AI Agents | Python, FastAPI, TypeScript, MySQL, Redis, Lua, SQS, LiteLLM*

- Built the **Campaign Manager** end to end: lead uploads, scheduling and outbound AI calling, supporting ∼40 clients.
- Scaled it to **5 lakh+ calls a day** and **700 concurrent calls** across all clients, with automatic retries.
- Reworked the **telephony layer**: SIP trunks, number-to-agent routing and support for a new calling provider.
- As a **forward-deployed engineer**, integrated our agents with edtech and retail clients' systems (**$200K+ ARR**).
- Cut **database load by 70%** by moving blocking calls to async, and added fair scheduling across lead queues.
- Built a queue orchestrator with per-agent **rate limiting**, so calls trigger reliably without overloading any agent.
- Built a **configurable chat widget** as a new channel, so clients could embed their chat agents on any website.

***Fermi AI** – AI Tutoring | Python, FastAPI, PostgreSQL, Celery, Redis, AWS, Terraform, Gemini*

- Built **usage-based billing** for AI tutors, charged per active minute across three products, **no double-charges**.
- Built a **targeting & A/B testing platform** so product teams can run experiments on user segments themselves.
- Launched **Scan & Ask** (snap a problem, get an AI tutor), live on India's **National Digital Library**, **98%** accuracy.
- Cut **database connections by 92%** (64 to 5) when traffic was about to cross the database's limit.
- Set up **deploys and monitoring** for all services, with dashboards, alerts and a **daily SLO report**.

Projects

GroupMailBox – Lead-capture & Email SaaS | *TypeScript, Chrome MV3, Express, BullMQ*

- Published a **Chrome extension** that captures Facebook Group leads, removes duplicates and saves them to Sheets.
- Built a **Gmail campaign engine** sending **100 emails/min**, with retries, daily caps and one-click unsubscribe.
- Added **bulk member-request review** for group admins, checking each approval and capped at **50 per session**.

Relay – Durable Workflow Engine | *Python, FastAPI, Postgres, OpenTelemetry*

- Built a **durable workflow engine** that records every step, so **1M** in-flight runs survive any crash.
- Chaos-tested it by killing workers mid-run across **1,000** workflows: exactly **1,000** charges, none doubled.
- Designed it to handle **1M workflows a day** on a single database, **5k transactions/s** at peak.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Lua
Backend: FastAPI, Node.js, Express.js, Celery, BullMQ, REST APIs, WebSockets, Microservices, Async Processing
Systems: Distributed Systems, System Design, Message Queues, Rate Limiting, Caching, A/B Testing, Observability
AI / LLM: LLM Integration, LiteLLM, MCP, LangGraph, AI Agents, Prompt Evaluation
Databases: PostgreSQL, MySQL, Redis, MongoDB, DynamoDB, ClickHouse
Cloud and DevOps: AWS (ECS, RDS, SQS, S3), Docker, Kubernetes, Terraform, CI/CD, Grafana, CloudWatch
Frontend: React, Next.js, HTML5, CSS

Achievements

- **Codeforces** Specialist (**1562**), **LeetCode** Knight (**1860**), **CodeChef** 4⋆ (**1884**); Codeforces Round 760 rank **38**.
- **Top 1600** of **400k+** teams in **Flipkart Grid 5.0**; **Top 20** of **2000+** students on **GeeksforGeeks** in college.
- General rank **9085** in **IIT JEE Advanced**.